Lab 1 - Advanced Machine Learning

Linear Regression

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General Informations

- **Assignment:** You will work **in pairs**.
- **Due date: Each assignment must be sent within 7 days of the lab session (by the next Thursday evening).**
  Send your notebook to `ludovic.trautmann@inria.fr` (only one submission per pair, remember to put both of your names on it).
- **Grading:** There are 6 lab sessions. The average grade will count for one third of your final grade.
  One part of the final exam will cover the material from the labs.

Lesson Recap

Linear model

We suppose that the relation between the input variable $x_1, \ldots, x_d$ and the output y is a linear relation

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_d x_d$$

Training

We use a training set of n examples $\mathbf{X} \in \mathbb{R}^{n \times (d+1)}$ with prediction $\mathbf{Y} \in \mathbb{R}^n$.

We aim to find the best $\beta_0, \ldots \beta_d$
$\rightarrow$ What does *best* mean ?

We aim to minimize an **objective function**, using a **well-suited algorithm**.

Least Squares

Objective

$$\underset{\beta \in \mathbb{R}^{d+1}}{\text{minimize}} \ \|\mathbf{X}\beta - \mathbf{Y}\|_2^2$$

Closed-form solution : $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$

Well suited-algorithm : Gradient descent

Effect : /

Ridge

Objective

$$\underset{\beta \in \mathbb{R}^{d+1}}{\text{minimize}} \ \|\mathbf{X}\beta - \mathbf{Y}\|_2^2 + \lambda \|\beta\|_2^2$$

Closed-form solution : $\hat{\beta} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}$

Well suited-algorithm : Gradient descent

Effect : Shrinkage

Lasso

Objective

$$\underset{\beta \in \mathbb{R}^{d+1}}{\text{minimize}} \ \|\mathbf{X}\beta - \mathbf{Y}\|_2^2 + \lambda \|\beta\|_1^2$$

Closed-form solution : Unavailable

Well suited-algorithm : Iterative Shrinkage-Thresholding Algorithm

Effect : Sparsity

Robust

Objective

$$\underset{\beta \in \mathbb{R}^{d+1}}{\text{minimize}} \sum_{i=1}^{n} \rho([\mathbf{X}\beta - \mathbf{Y}]_i)$$

Closed-form solution : Unavailable

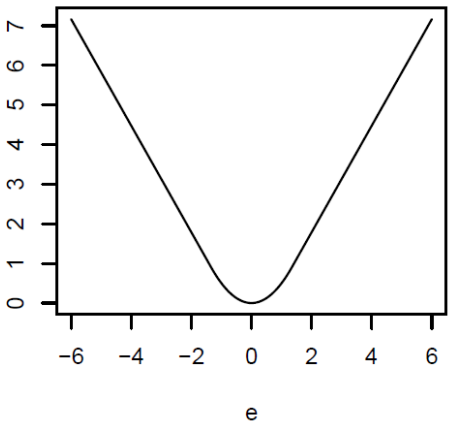
Well suited-algorithm : Iteratively Reweighted Least Square (IRLS)

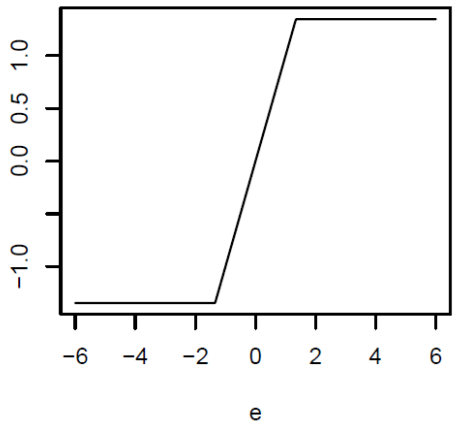
Effect : Robustness

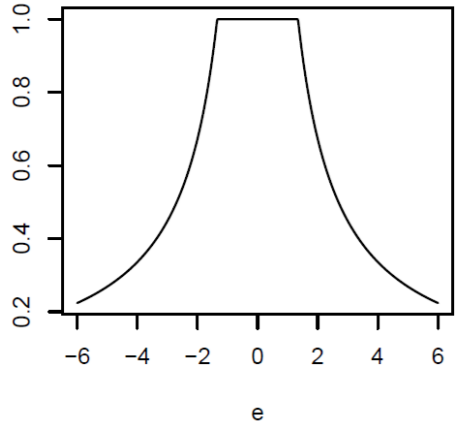
Example of potential functions

Huber potential

$$\rho(e) = \begin{cases} \dfrac{e^2}{2} & \text{if } |e| \leq \delta \\ \delta|e| - \dfrac{\delta^2}{2} & \text{if } |e| > \delta \end{cases}$$





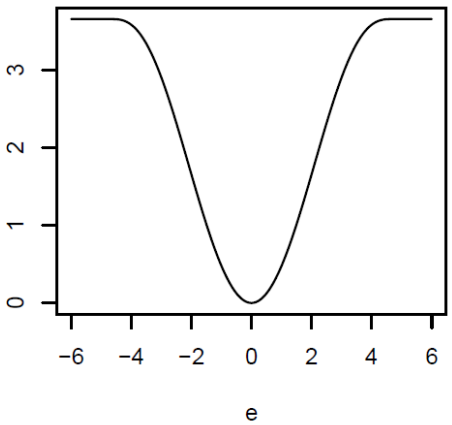


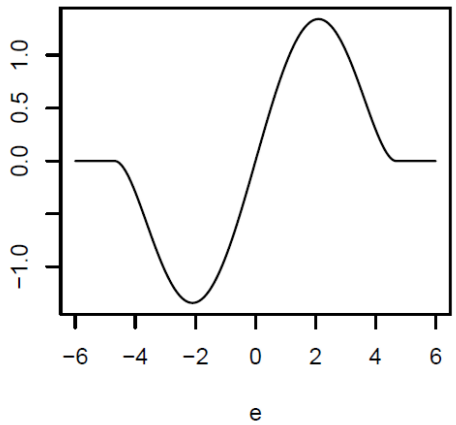
$$\rho(e) \quad - \quad \dot{\rho}(e) \quad - \quad \frac{\dot{\rho}(e)}{e}$$

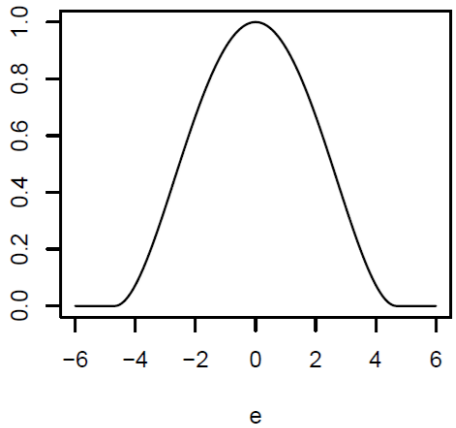
Example of potential functions

Bisquare potential

$$\rho(e) = \begin{cases} \frac{\delta^2}{2}\left(1 - \left(1 - \frac{e^2}{\delta^2}\right)^3\right) & \text{if } |e| \leq \delta \\ \frac{\delta^2}{6} & \text{if } |e| > \delta \end{cases}$$







$$\rho(e) \quad - \quad \dot{\rho}(e) \quad - \quad \frac{\dot{\rho}(e)}{e}$$